

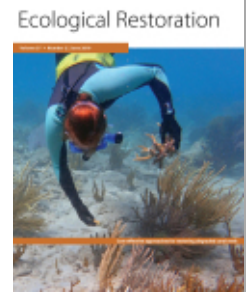


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Ecosystems

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Restoration of Ecosystem Function by Soil Surface Inoculation with Biocrust in Mesic and Xeric Alpine Ecosystems

Annie-Claude Letendre, Darwyn S. Coxson and Katherine J. Stewart

ABSTRACT

Biocrusts are soil surface assemblages of lichens, bacteria, fungi, and bryophytes that support key ecosystem processes in alpine environments. Their role in carbon and nitrogen cycling is critical to soil pedogenesis. Establishment of biocrusts after disturbance events is an important first step in restoring ecosystem function. However, our understanding of active restoration of alpine biocrusts, as well as how environmental factors affect ecosystem processes associated with early succession, is poor. In a field experiment at two alpine sites (a xeric and a mesic site), we monitored microclimatic conditions and assessed function of mature biocrusts. We also examined the influence of inoculation with mature biocrusts and fertilization on the recovery of biocrusts in disturbed alpine soils. Biocrust cover and soil surface nitrogen fixation were enhanced by inoculation after 12 weeks at the mesic site. However, at the xeric site biocrust cover increased while nitrogen fixation declined. Restoration of dry and cold alpine environments presents many challenges and recovery of ecosystem services may require additional restoration techniques and longer timelines. Improved understanding of the community composition of alpine biocrusts and their developmental trajectories over time will assist with effectively using biocrusts to promote recovery of ecosystem processes in disturbed alpine environments.

Keywords: biological soil crusts, nitrogen fixation, extracellular polysaccharides, inoculation

🌿 Restoration Recap 🌿

- Inoculation of disturbed alpine environments with biocrusts may be an effective restoration technique under mesic climate conditions, however successful establishment of inoculum under cold xeric environments may pose challenges.
- Establishment of biocrust cover in restored cold xeric environments may not be indicative of recovery of ecological function, such as soil surface nitrogen fixation.
- In cold xeric climates, restoration protocols that increase surface moisture retention are likely needed for successfully restoration of biocrust communities.
- Alpine biocrusts appear to play a significant role in nitrogen and carbon input and restoration of these communities following disturbance may support ecosystem recovery.

Anthropogenic disturbances can have severe impacts on alpine vegetation and soils (Bayramov et al. 2012, Desserud and Naeth 2013, Bulot et al. 2014) causing decreases in plant and microbial diversity and alteration of soil organic carbon and nitrogen cycling processes. (Buttars et al. 1998, Capers and Taylor 2014, Coleine et al. 2015). The recovery of alpine environments after disturbances is commonly limited by nutrient-poor soils and short,

cool growing seasons (Chambers et al. 1990, Scherrer and Pickering, 2006, Bayramov et al. 2012). Biocrust inoculation may provide a tool for restoration of disturbed alpine soils. Biocrusts are soil surface communities formed by a diversity of organisms, including algae, lichens, mosses, liverworts, and cyanobacteria (Peer et al. 2010, Steven et al. 2015, Čapková et al. 2016). Although biocrusts are sensitive to disturbances (Belnap and Eldridge 2001, Peer et al. 2010, Kuske et al. 2012, Steven et al. 2015) they are well-adapted to harsh alpine growing conditions (Čapková et al. 2016) and can facilitate the recovery of ecosystem services (Jefries et al. 1992, Evans and Belnap 1995, Gretarsdottir et al. 2004, Ferrenberg et al. 2017).

Established biocrust communities play a key role in soil nitrogen and carbon cycling through processes such as nitrogen fixation and the production of extracellular polysaccharides (EPS) (Maestre et al. 2011, Bowker et al. 2014, Delgado-Baquerizo et al. 2016). Nitrogen typically limits primary production during early successional stages of alpine plant communities (Schmidt et al. 2012) and the cyanobacteria and cyanolichens that compose alpine biocrusts play a fundamental role in fixing nitrogen and making it available to plants (Aanderud et al. 2018). Additionally, biocrusts may play a significant role in the cycling of trace gases such as CO₂ (Zaady et al. 2000, Stark et al., 2011), as well as retaining nutrients such as sodium, magnesium, calcium, iron, nickel, copper, and zinc (Bowker et al. 2008, Yu et al. 2014). EPS compounds secreted by cyanobacteria and green algae in biocrusts form an envelope surrounding the cells that bind soil particles and retain water (Li et al. 2009). Therefore, soils colonized by biocrusts have increased stability and are more resistant to erosion (Belnap, 1995, Zaady et al., 2000, Bowker et al., 2008). EPS attract and absorb water from their environment, which makes them particularly important in biocrust communities where bryophytes and other poikilohydrous organisms require externally provided water for growth and reproduction (Grote et al., 2010, de Guevara et al., 2014). Furthermore, biocrusts influence ecosystem hydrology through altering runoff and infiltration of water. The microtopography created by biocrusts also provides habitat for soil microbes and surface roughness to trap wind-blown seed. Through these ecosystem functions, biocrusts modify soils to affect vascular plant seed dispersal, germination, establishment, survival, and nutritional status (Belnap, 2002, Hawkes, 2004, Maestre et al. 2006, Bowker et al. 2007, Bowker et al. 2011, Bu et al. 2013).

Given the importance of biocrust establishment for the restoration of soil surface properties (e.g., soil stabilization, nitrogen and carbon enrichment) following disturbance, the artificial enhancement of biocrust establishment on disturbed sites has been the subject of a number of studies, using techniques such as cultivation and inoculation (Bowker 2007, Xu et al. 2008, Liu et al. 2012, Stewart and Siciliano, 2015). While previous studies have examined the efficacy of inoculation for establishing biocrusts in grassland and desert sites (Belnap 1993, Maestre et al. 2006, Steven et al. 2015), to our knowledge none have investigated the effects of assisted inoculation on the establishment and function of alpine biocrusts.

We address this knowledge gap in a study of alpine biocrusts in two contrasting sites: the mesic coastal mountain ranges of northwestern British Columbia (Terrace, British Columbia, Canada), and the xeric interior mountain ranges of the southern Yukon (Whitehorse, Yukon Territory, Canada). Our aims were to characterize differences in the xeric and mesic alpine soil crust environments, and to assess differences in biocrust establishment and ecological

function following restoration (i.e., inoculation) at the two sites. At each location, we examined the cover and function of biocrusts after disturbance in controlled experimental plots. Experimental treatments included plots with and without biocrust inoculation, and plots with and without fertilizer addition. Use of fertilizers is common in alpine restoration (Onipchenko et al. 2012, Ma et al. 2014) and we included a fertilization treatment to examine how this practice may influence biocrust inoculation. Soil surface microclimate was monitored and nitrogen fixation was measured at each site to provide a better understanding of the operating conditions under which physiological activity occurs in these contrasting biocrust communities. Our main objectives were to 1) determine the operating environment of natural biocrusts associated with contrasting coastal and continental mountain ranges, 2) assess the restoration potential of these alpine biocrust communities, and 3) evaluate the influence of biocrust inoculation treatments on nitrogen fixation, extracellular polysaccharide content, soil mineralizable nitrogen and dissolved organic carbon, as indicators of soil ecological function.

Methods

Study Areas

Coastal alpine plots (mesic) were situated on Andesite Peak (54°13.868' N, 128°01.499' W, 1640 m elevation) in the Coast Mountain range near Terrace, British Columbia. Predicted mean annual temperature and precipitation on Andesite (1640 m a.s.l.) is -1°C and 2019 mm respectively (Environment Canada 2017). Alpine tundra vegetation on Andesite was dominated by the shrub *Cassiope mertensiana* (white mountain heather) and *Stereocaulon alpinum* (snow lichens) on ridge top habitats, while herbs *Saxifraga tolmiei* (Tolmie's alpine saxifrage) and *Ranunculus cooleyae* (Cooley's buttercup) were common in depressions. This site is characterized by Podzolic and Regosolic soils (Pojar and Stewart 1991), with substrates of volcanic origin (meta-basalts and meta-andesites).

Continental alpine plots (xeric) were located on Mt McIntyre (60°63.797' N, 135°16.711' W, 1370 m elevation) on the Teslin Plateau near Whitehorse, Yukon, Canada. Mean annual precipitation in Whitehorse (706 m a.s.l.) is 262 mm, with roughly half falling as snowfall (Strong 2013). Vegetation on Mt McIntyre was dominated by shrubs including *Salix* spp. (willow), *Vaccinium* spp. (dwarf heath shrubs), and *Dryas* spp. (cushion-forming evergreen shrubs). Herbs including *Chamerion angustifolium* (fireweed) and *Anemone* spp. (anemones), and lichens including *Stereocaulon* spp. (snow lichens), and *Cladina* spp., *Cladonia* spp. (reindeer moss) were common. Located in the Mid-Cordilleran Alpine ecoclimatic region of Yukon, the area is characterized by sedimentary rocks (clastic/limestone) and Brunisolic soils (Strong 2013).

At each site five 60-m transects located across representative slope and aspect were used to survey biocrust cover. Total percent cover (i.e., presence/absence) was assessed by ocular estimation at 5-m intervals along the transect in 1-m² plots. Biocrust cover was 22.4 ($\pm$ 2.8 SE) % on Andesite Peak, and 23.0 ($\pm$ 3.4 SE) % on Mt McIntyre. Representative samples of the biocrust community were collected for species identification. Species were not quantified within these samples and provide only a description of the mature biocrust community at each site. On Andesite Peak, 40 lichen, 12 liverwort and 17 moss species were identified in mature biocrusts. *Amygdalaria panaeola*, *Psoroma tenue* var. *boreale*, *Solorina crocea*, as well as *Stereocaulon alpinum*, *botryosum*, and *vesuvianum* formed the primary nitrogen fixing cyanolichen biocrust community. On Mt McIntyre, 57 lichen, 3 liverwort and 21 moss species were identified in mature biocrust. Nitrogen-fixing lichens were *Fuscopannaria praetermissa*, *Nephroma arcticum*, *Peltigera* spp., *Psoroma tenue* var. *boreale*, and *Stereocaulon alpinum* and *tomentosum*. Two nitrogen-fixing lichens were found at both sites (*Psoroma tenue* var. *boreale* and *Stereocaulon alpinum*), as well as 11 other lichen, 2 liverwort (*Anastrophyllum minutum* var. *minutum* and *Tritomaria quinqueidentata*) and 3 moss species (*Dicranoweisia crispula*, *Pogonatum contortum*, *Polytrichum piliferum*).

Microclimate Monitoring of Natural Biocrust

At each site, a microclimate monitoring station was installed. At Andesite, a CR1000 datalogger (Campbell Scientific Inc., Logan, UT) was coupled to a AM416 multiplexer (Campbell Scientific Inc., Logan, UT), fitted with a HPM45 temperature/relative humidity probe (Campbell Scientific Inc., Logan, UT), two EC-5 soil moisture/temperature probes (Decagon Devices Inc., Pullman, WA), and two LiCor Photosynthetically Active Radiation (PAR) sensors (LiCor Inc., Lincoln, Nebraska). Twelve Omega 0.001 mm copper/constantan (Cu/Cn) thermocouples (Omega Engineering, Stamford, Connecticut) and custom-made impedance clips (Coxson 1991) were attached to the surface layer of mature biocrusts. At Mt McIntyre, a CR23 datalogger (Campbell Scientific Inc., Logan, UT) and a CR1000 datalogger (Campbell Scientific Inc., Logan, UT) coupled to an AM25T multiplexer (Campbell Scientific Inc., Logan, UT) were fitted with the same probes and sensors. The Andesite microclimate monitoring station was in an open alpine-tundra habitat, with no shrub cover. The Mt McIntyre microclimate monitoring station was in a sparse shrub-alpine tundra complex. Consequently, soil surface probes at Mt McIntyre were partially shaded from direct solar input.

Field Experimental Design

Four treatments were applied to artificially disturbed plots in a randomized block design with nine and ten blocks established at Andesite and Mt McIntyre, respectively.

Each block was located in an area with uniform topography where biocrusts were present and vegetation sparse. Each treatment was applied within a 1-m² area, randomly assigned within each block with a buffer of 0.5 m between treatment plots. The four treatments were biocrusts with fertilizer (BF), biocrusts only (B), fertilizer only (F), and a control (C) with no biocrusts or fertilizer applied. Fertilizer was a N:P:K (as NH₄NO₃:P₂O₅:K₂O) of 19:19:19 applied at a rate of 110 kg/ha, equivalent to ~21 kg/ha of each nutrient. This fertilization is an industry standard commonly used on sites with moderate to high N, P, K levels in the Yukon Territory, Canada (Matheus and Omtzigt 2013). The fertilizer was in granular form and to assist with even application, fertilizer granules were pulverized with a mortar and pestle. The pulverized fertilizer was applied by spreading the powder evenly over each fertilizer plot.

Biocrusts used to prepare the inoculation mixture for restoration plots were collected within areas delineated for disturbance treatment plots. All biocrusts and immediately underlying soil within each 1-m² treatment plot were cut from the soil surface with a knife to a depth of 2.5 cm. Vegetation in treatment plots was removed. Collected biocrusts were sieved (4.5 mm mesh) and homogenized. Once biocrusts were removed from plots, the soil surface was disturbed by raking and cultivating with a handheld garden cultivator to a depth of 3 cm. Homogenized biocrusts were then spread evenly across 1-m² plots for the B and BF treatments (n = 18 and n = 20 for Andesite and Mt McIntyre, respectively) at a rate of 10% surface area (i.e., 0.1 m² per plot). A previous study demonstrated successful establishment with a 10% inoculation (Stewart and Siciliano, 2015) and a preliminary trial showed recovery of the community with 10% inoculation (as compared to 25% inoculation), which allows for efficient use of mature biocrust materials. The F and C treatments (n = 18 and n = 20 for Andesite and Mt McIntyre, respectively) did not receive biocrust applications.

Soil surface samples were collected for analysis of biocrust recovery within experimental plots at weeks 0, 6, and 12 following inoculation. A composite sample (64 cm², 0.5–1 cm depth) was collected from each treatment plot. Sampling locations within each plot were determined through random numbers associated with an overlaid grid. Total percent cover of biocrust was estimated by eye at each sampling location. Samples were not taken from within 10 cm of the edge of the plot. Samples were held in the dark at 2°C prior to conducting Acetylene Reduction Assays (ARAs).

Natural Biocrust Sampling

Naturally occurring mature biocrusts from adjacent undisturbed alpine tundra were sampled to provide a reference for recovery of nitrogen fixation in the experimental plots at week 12. At each site ten samples of undisturbed mature biocrust (0.5 cm to 2 cm depth) were collected from eight

locations representative of biocrust composition at each site ($n = 80$ per site).

Acetylene-Reduction Assays

The nitrogen fixation potential of soil surface samples (from mature biocrust and experimental restoration plots) was assessed using acetylene reduction assays (ARAs) (Stewart et al. 1967). Immediately prior to conducting ARAs biocrusts were given a 24 hour (h) pre-treatment incubation consisting of 16 h light ($250 \mu\text{mol m}^{-2} \text{s}^{-1}$)/8 h dark, at 20 and 10°C respectively. Samples were then placed into 250 mL glass canning jars with serum stopper lids. Samples were misted with 1 mL of deionized water prior to closure of jars and injection of 10% acetylene by volume. Jars were incubated 4 h at $150 \mu\text{mol m}^{-2} \text{s}^{-1}$ PAR and 20°C. After incubation a 4-mL gas sample was removed from each incubation jar for analysis of ethylene concentration. Each set of samples was processed with two control samples: i) a biocrust sample not injected with acetylene, to assess the degree to which the natural evolution of ethylene occurred, and ii) a jar with no biocrusts that was injected with acetylene to assess ethylene contamination, if any, in the acetylene used. Gas samples were injected into a gas chromatograph (SRI 8610A, Wennick Scientific Corporation, Ottawa, ON, Canada) fitted with a Porapak column (Alltech Canada, Guelph, ON, Canada) and a flame ionization detector for detection of ethylene. Hydrogen, used as the carrier gas, was held at a constant pressure of 32 psi while column temperature was held at 65°C. After ARAs were completed, samples from week 0 and 12 collections were homogenized and kept frozen at -20°C, for subsequent extracellular polysaccharide analysis.

Although a theoretical conversion ratio of 3:1 is commonly used for the ratio of ethylene reduced to nitrogen fixed, we have left our data as μmol ethylene because of the wide range of conversion ratios reported for terrestrial species (0.1–6) along with their strong dependency on environmental conditions (Evans and Lange 2001, Barger et al. 2016). See the discussion for further limitations of ARAs.

Extracellular polysaccharide content

In order to assess carbon inputs and soil stability, extracellular polysaccharide (EPS) content was measured on samples from experimental plots taken at weeks 0 and 12 ($n = 76$ and 80 for Andesite and Mt McIntyre, respectively). EPS content was determined using the phenol-sulfuric acid method (Crayton 1982, Masuko et al. 2005). Reactions were transferred to a 96-well flat-bottom microplate, with standards and template control carried out in triplicate. EPS content was measured as glucose concentration quantified at 480, 486, and 490 nm using a microplate reader (Varioskan, Thermo Fisher Scientific, Waltham, MA).

Soil properties

Dissolved organic carbon and mineralizable nitrogen were measured on soil samples collected at week 12 (composite samples) from each plot ($n = 36$ and $n = 40$ for Andesite and Mt McIntyre, respectively). Soil was sampled under biocrusts to a depth of 3 cm. Dissolved organic carbon was measured from filtered ($0.45 \mu\text{m}$) soil extracts (1:10 soil to water ratio shaken for 1 h on a reciprocating shaker) using a Total Organic Carbon analyzer (Formacs HT, Skalar Analytical B.V., Breda, Netherlands). Mineralizable nitrogen was measured by the anaerobic incubation method (Waring and Bremner 1964). Soil samples were incubated under waterlogged conditions for two weeks at 30°C. Ammonium produced was subsequently measured from a soil extract (1:10 soil to 1N KCl ratio shaken for 2 h on a reciprocating shaker) with a discrete analyzer using the phenate method for colorimetric determinations (Solorzano 1969; SmartChem 200, Unity Scientific, Brookfield, CT).

Data Analysis

Linear mixed models (LMM) were used for the analysis of ARAs and EPS content. Since samples were collected from the same plots at multiple time points, pseudoreplication within plots was avoided by including plot number as a random effect. LMM included plot number as a random effect, week, and measurement (i.e., ARAs and EPS content) as fixed effects, and the interaction of week and measurement as fixed effects. To meet assumptions of normality (Shapiro-Wilk test) and homoscedasticity (Bartlett's test) all measurements except dissolved organic carbon content were log-transformed. Acetylene reduction, and EPS content data were analysed using R package lme4 (Bates et al. 2014) to generate LMM fitted by restricted maximum likelihood (REML) and the R package lmerTest (Kuznetsova et al. 2013) to calculate degrees of freedom through the Satterthwait approximation. Post-hoc testing was carried out with Tukey HSD from R package multcomp (Hothorn et al. 2008). Soil properties and biocrust percent cover were analyzed using ANOVA and Tukey HSD post-hoc tests. All statistical analyses were conducted in R version 3.3.1 (R Core Team, 2014).

Results

Characterization of Alpine Biocrust Operating Environment

At Andesite, cumulative daily photosynthetically active radiation (PAR) showed a strong seasonal pattern, declining from over $55 \text{ mol m}^{-2} \text{ day}^{-1}$ in early summer to less than $25 \text{ mol m}^{-2} \text{ day}^{-1}$ by mid-September (Figure 1). Mt McIntyre had a similar pattern of decline, from high early summer PAR values ($25 \text{ mol m}^{-2} \text{ day}^{-1}$) to low early

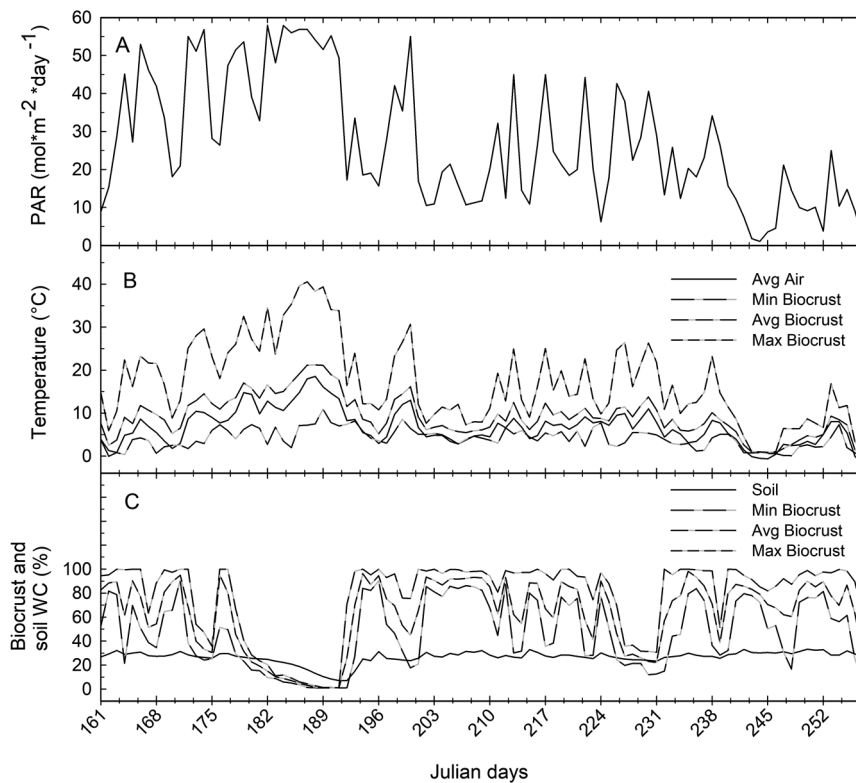


Figure 1. Microclimate data from mature biocrust monitoring station on Andesite Peak over the duration of the experiment, expressed in Julian days. Photo-synthetically active radiation (PAR; $\text{mol m}^{-2} \text{ day}^{-1}$) (A). Average air temperature ($^{\circ}\text{C}$; solid line) and the minimum, average, and maximum biocrust temperatures ($^{\circ}\text{C}$; long, medium, and short-dashed lines respectively) (B). Average soil volumetric water content (%; solid line), as well as the minimum, average, and maximum biocrust relative water content (%; long, medium, and short dashed lines respectively) (C).

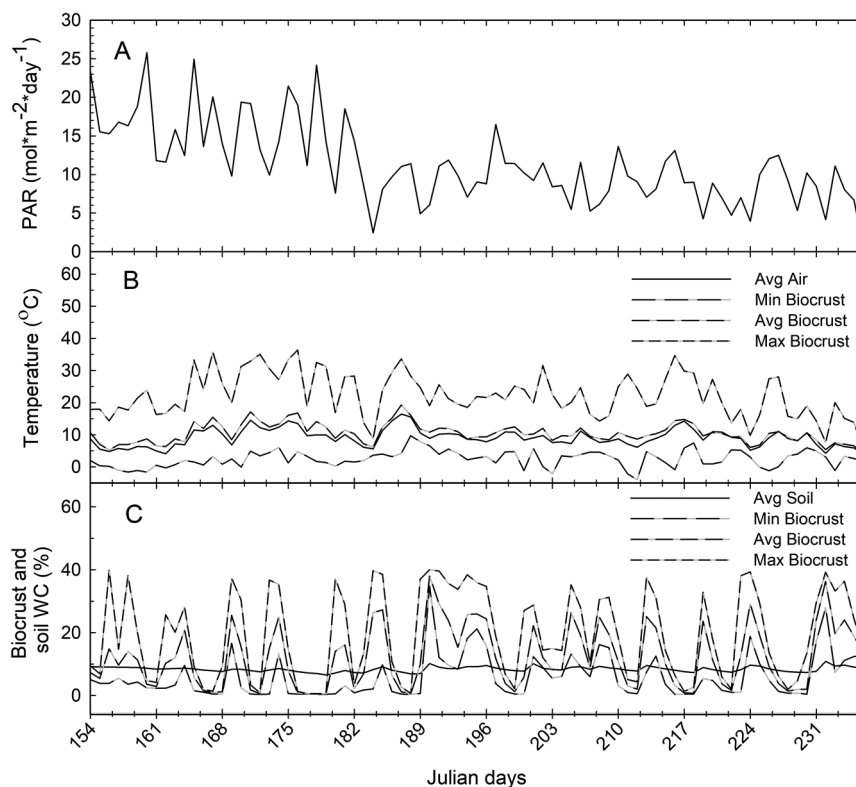


Figure 2. Microclimate data from mature biocrust monitoring station on Mt McIntyre over the duration of the experiment, expressed in Julian days. Photo-synthetically active radiation (PAR; $\text{mol m}^{-2} \text{ day}^{-1}$) (A). Average air temperature ($^{\circ}\text{C}$; solid line) and the minimum, average, and maximum biocrust temperatures ($^{\circ}\text{C}$; long, medium, and short-dashed lines respectively) (B). Average soil volumetric water content (%; solid line) as well as the minimum, average, and maximum biocrust relative water content (%; long, medium, and short-dashed lines respectively) (C).

fall ($12 \text{ mol m}^{-2} \text{ day}^{-1}$) (Figure 2). Overall, lower solar input was observed at the Mt McIntyre site.

Mean daily biocrust temperatures were consistently higher than mean daily air temperatures at both Andesite and Mt McIntyre (on average by 2.5 and 1.4°C , respectively). At Andesite, biocrust temperatures reached a maximum of

40.6°C on 6 July and a minimum of -1.6°C on 14 September. At Mt McIntyre, maximum biocrust temperature was 36.3°C on 25 June and minimum biocrust temperature -4.0°C on 31 July. Biocrust temperatures below 0°C were recorded only once at Andesite. In contrast, at Mt McIntyre sub-zero temperatures were observed on 16 days over the summer.

Table 1. ANOVA on mixed models of function parameters and linear models of soil properties and biocrust cover of experimental plots at Andesite and Mt. McIntyre (Site) sampled 0, 6, 12 weeks or 0, 12 weeks following inoculation (Time). TR = treatment (control, fertilizer only, biocrust only, and biocrust + fertilizer). Min N = mineralizable nitrogen and DOC = dissolved organic carbon. Fvalues are not available for the factor Time, for Cover, DOC and Min N that were only measured at 12 weeks. *significant values ($p < 0.05$).

Factor	Function Parameter				Soil Properties				
	df	NA	df	EPS	df	DOC	Min N	df	Cover
		F-value (<i>p</i> -value)		F-value (<i>p</i> -value)		F-value (<i>p</i> -value)			F-value (<i>p</i> -value)
TR	3	0.96 (0.42)	3	3.7 (0.02)	3	40 (< 0.01)*	2.0 (0.12)	3	25 (< 0.01)*
Time	2	16 (< 0.01)*	1	0.70 (0.41)	1	—	—	—	—
Site	1	40 (< 0.01)*	1	76 (< 0.01)*	1	157 (< 0.01)*	35 (< 0.01)*	1	1.6 (0.20)
TR * Time	6	1.2 (0.32)	3	0.50 (0.68)	3	—	—	—	—
TR * Site	3	0.54 (0.66)	3	2.9 (0.04)	3	33 (< 0.01)*	0.7 (0.55)	—	2.6 (0.06)
Time * Site	2	82 (< 0.01)*	1	1.7 (0.19)	1	—	—	—	—
TR * Time * Site	6	0.94 (0.47)	3	0.28 (0.84)	3	—	—	—	—

The total duration of hydration episodes, defined as biocrust relative water content (RWC) > 25%, for biocrust at Mt McIntyre (907 h) was considerably shorter than that of Andesite (1589 h), although the mean temperature of hydrated crusts was similar (9.4 and 12.1°C, respectively). Mean volumetric soil water content was 26.6% at Andesite and 21.0% at Mt McIntyre. RWC averaged 62.9% at Andesite and 30.6% at Mt McIntyre. Overall, biocrusts and surface soils (5-cm depth) were drier at Mt McIntyre than at Andesite.

Recovery of Inoculated Biocrust and Associated Ecosystem Functions

Treatment plots receiving biocrust inoculation (B and BF) had significantly higher biocrust cover after 12 weeks ($33\% \pm 3$ and $32\% \pm 2$ at Andesite, and $22\% \pm 2$ and $20\% \pm 3$ at Mt McIntyre; Table 1, Figure 3). Addition of fertilizer with the inoculant did not significantly increase cover of biocrusts at either site. Similarly, fertilization alone did not significantly increase natural recolonization. Fertilized plots (F) and control soils (C) had a $3\% \pm 1$ and $10\% \pm 3$ biocrust cover at Andesite and $8\% \pm 1$ and $7\% \pm 0.1$ cover at Mt McIntyre. There was a significant interaction between site and treatment (Table 1). Inoculated plots (B and BF) had higher cover at Andesite compared to Mt McIntyre; however, biocrust cover of uninoculated plots (C and F) was not significantly different between sites (Figure 3).

Nitrogenase activity (NA) measured by acetylene reduction assays changed significantly over time at both sites; however, opposing trends were observed (Table 1, Figure 4). At Mt McIntyre, NA declined significantly from week 0 to 12, whereas at Andesite, NA increased significantly from week 0 to 12 (Figure 4). NA was significantly higher at Andesite compared to Mt McIntyre in the restoration plots. In addition, NA was also higher in the mature biocrust measured at week 12 at Andesite (9.0 ± 3.9 μmol ethylene/ m^2/h) compared with Mt McIntyre (2.7 ± 3.7 μmol

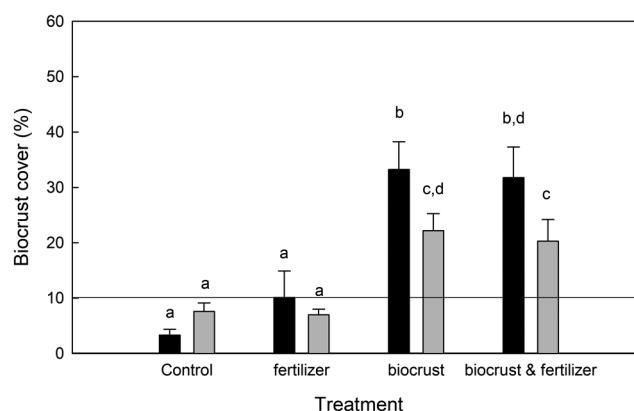


Figure 3. Biocrust cover 12 weeks following inoculation. The horizontal line represents the initial inoculation level of 10% biocrust (surface area based). At both Andesite (black bars) and Mt McIntyre (grey bars), plots inoculated with biocrust (biocrust and biocrust & fertilizer) had a significantly higher biocrust cover than uninoculated plots (control and fertilizer). Different letters indicate significant differences. Bars represent mean with standard error.

ethylene/ m^2/h). Nitrogen fixation rates in the restoration plots were slightly below rates for mature crusts at both sites in week 12 (~ 2 μmol ethylene/ m^2/hr at each site). Higher NA was detected in the restoration plots at week 0 than in the mature crust at week 12 at Mt McIntyre, which may reflect a seasonal pattern of declining NA.

Extracellular polysaccharide (EPS) content was significantly higher in Mt McIntyre samples (Figure 5, Table 1). There were no significant treatment or time differences at Andesite. However, at Mt McIntyre, 12 weeks following inoculation, the fertilizer treatment (F) had significantly lower EPS (Figure 5, 76 ± 19 μg of glucose per g of biocrust) compared to all other treatments (Figure 5, B = 250 ± 92 , BF = 217 ± 49 , C = 170 ± 42 μg of glucose per g of biocrust, respectively).

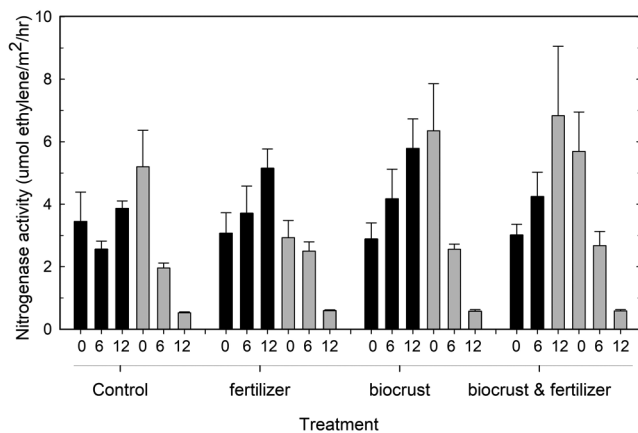


Figure 4. Acetylene Reduced 0, 6, and 12 weeks following inoculation with biocrust at Andesite (black bars) and at Mt McIntyre (grey bars). Acetylene reduction as a measure of nitrogen fixation increased at Andesite and decreased at Mt McIntyre over the 12 weeks. Bars represent mean with standard error.

Twelve weeks after inoculation, soil mineralizable nitrogen measured in the top three cm under biocrusts was significantly lower at Andesite than Mt McIntyre, but not significantly different between treatments at either site (Figure 6, Table 1). Conversely, dissolved organic carbon (DOC) was significantly higher at Andesite than at Mt McIntyre (Figure 6, Table 1). At Andesite, DOC was significantly higher in inoculated plots ($B = 75 \pm 4.9$ and $BF = 122 \pm 14$ ppm) than in uninoculated plots ($C = 9.2 \pm 1.3$ and $F = 45 \pm 5.8$ ppm). Higher DOC was observed in both inoculated ($BF > B$) and uninoculated plots ($F > C$) that received the fertilizer treatment respectively.

Discussion

Characterization of Alpine Biocrust Operating Environment

There was a clear contrast between soil surface microclimate at Mt McIntyre and the Andesite Peak. Frequent precipitation during the summer months at Andesite Peak are typical of coastal climate and prolonged the duration of biocrust hydration. In contrast, at the more xeric continental site, biocrust moisture availability at Mt McIntyre was derived mainly from spring snowmelt, with biocrusts becoming progressively desiccated as the summer progressed. Increased moisture availability in alpine and arctic biocrusts is correlated with an increase in ARA, a measure of nitrogen fixation potential (Chapin et al. 1991, Zielke et al. 2002, Zielke et al. 2005, Gavazov et al. 2010, Patova et al. 2016). This is similar to biocrusts in drylands where moisture is known to strongly influence nitrogen fixation (Belnap 1995, Maestre et al. 2006, Antoninka et al. 2015). The relatively constant moisture availability at Andesite provides a favourable environment for establishment and

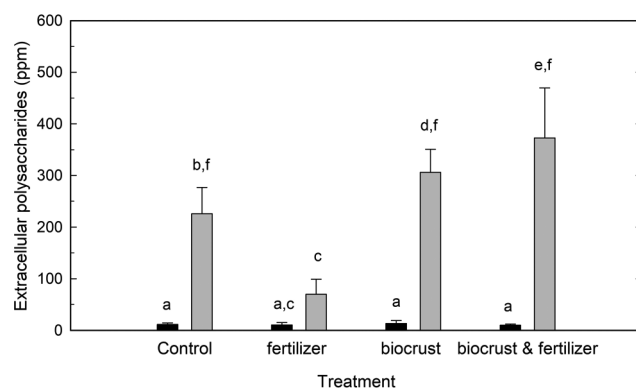


Figure 5. Extracellular polysaccharide content (EPS) of soil surface 12 weeks following inoculation at Andesite (black bars) and at Mt McIntyre (grey bars). Overall, EPS was significantly lower at Andesite than at Mt McIntyre. Different letters indicate significant differences. Bars represent means with standard error.

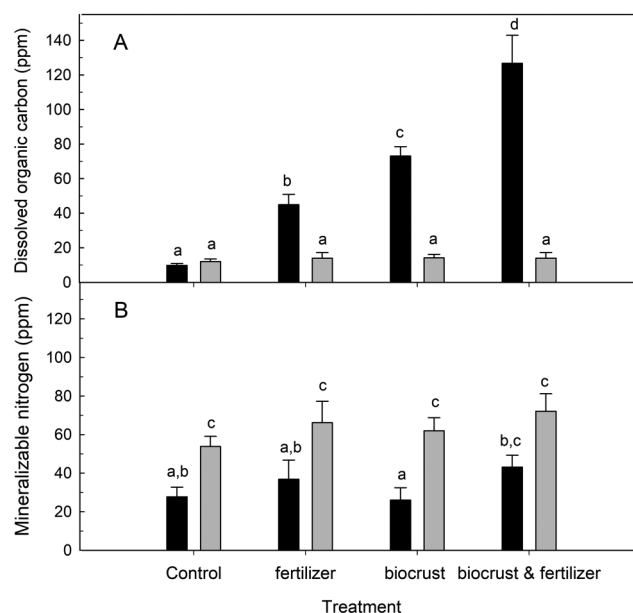


Figure 6. Soil properties 12 weeks following inoculation at Andesite (black bars) and at Mt McIntyre (grey bars). Dissolved organic carbon (A) was higher in inoculated plots (biocrust and biocrust & fertilizer) at Andesite. Mineralizable nitrogen (B) was higher at Mt McIntyre than at Andesite but not significantly different between treatments. Different letters indicate significant differences. Bars represent means with standard error.

growth of nitrogen-fixing species, reflected in optimal nitrogenase activity in mature crust being approximately three times higher at Andesite. Understanding the different operating conditions of these alpine sites has important implications for restoration practice. Xeric alpine sites that lack moisture over the growing season may require management of surface moisture, especially during the first season following inoculation. In addition, while Andesite

peak experienced only a single freezing event at the soil surface over the growing season, at Mt McIntyre 16 freezing events were observed suggesting that colder alpine environments may have a reduced timeline for restoration and harsher conditions that may reduce inoculum establishment.

Contrary to expectations that cumulative PAR exposure would be higher on Mt McIntyre due to its latitude and more cloud-free days, PAR exposure was higher for Andesite crusts. This reflects the shading of biocrusts by the shrubs (e.g., *Betula glandulosa* and *Salix* spp.) on Mt McIntyre. In contrast, dwarf shrubs (e.g., *Cassiope mertensiana*) at Andesite were closely appressed to the soil surface. Light saturation occurs at relatively low light levels in alpine and arctic biocrusts. Patova et al. (2016) found no significant increase in ARA rates at PAR values above 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$, and Zielke et al. (2002), found that biocrust ARA was saturated above 140 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Therefore, light is unlikely to be a limiting factor for biocrust activity at these alpine sites, whereas surface moisture may be critical.

Recovery of Inoculated Biocrust and Associated Ecosystem Functions

Mt McIntyre and Andesite had similar biocrust cover in experimental plots 12 weeks after inoculation (25–35%); however, we observed significantly greater recovery of biocrust function (i.e., nitrogen fixation and EPS) at Andesite. The declining pattern in acetylene-reduction over time at Mt McIntyre may reflect a seasonal trend, since we also observed lower NA in mature crust at week 12. Given the more northerly location of the Mt McIntyre site a shorter growing season may be expected, and this was observed in a greater number of days biocrust temperature was below 0°C at Mt McIntyre (i.e., 16 versus 1). Additionally, biocrust at the Mt McIntyre site underwent an uncharacteristically dry period at the beginning of our experiment. Normal local precipitation in June is 32 mm, but in June 2015 only 16 mm was received (Environment Canada 2017). This highlights the importance of water availability in restoration of montane environments, which is consistent with drylands where biocrust succession and functions are limited by moisture (Zaady and Offer 2010). In xeric cold climates, different restoration protocols that increase surface moisture retention are likely needed for successfully restoration of biocrusts.

Relatively few studies have quantified nitrogen fixation by biocrusts across alpine landscapes. In a study of cyanobacteria dominated biocrust of subpolar Ural Mountains, Patova et al. (2016) reported hourly rates of acetylene reduction between 0.5 and 1.76 mg of ethylene/m² (18–62 μmol of ethylene/m²). This was three to ten times higher than the rates from restored biocrusts on Andesite, though only twice that of mature crusts from Andesite incubated under optimal conditions. At high altitude nival and subnival sites in the Himalayan Mountains, nitrogen-fixation rates

ranged from 0.3 to 5 nmol of ethylene/g/day (Janatkova et al. 2013). In a high alpine ecosystem in the Austrian Alps, soils with biocrusts had twice as much total nitrogen as those without (Peer et al. 2010). Although exact calculation of nitrogen inputs by alpine biocrusts warrants more detailed investigations, our study supports the significance of nitrogen fixation by biocrusts in alpine environments.

Our measurement of nitrogen fixation via ARAs has limitations. There is high variability in ARA rates reported for biocrusts primarily due to ARAs acting as an indirect measurement of NA with conversion ratios that can vary from 0.022 to 3.49 (Barger et al. 2016). The deviation in these conversion ratios can be attributed to several factors including the higher solubility and therefore availability of acetylene in water compared with N₂, acetylene adsorption onto soil colloids, diffusion barriers, and toxicity to microbes (Barger et al. 2016). Calibration of ARAs with ¹⁵N₂ or direct use of ¹⁵N₂ is needed to determine absolute rates of NA.

Compared with restoration efforts of biocrust in drylands the establishment and recovery of ecosystem function was relatively rapid in these alpine environments. In a recent study inoculated biocrust material failed to develop in a dryland sandy site 10 years following inoculation (Chandler et al. 2018). A lateral expansion of approximately 1–3 cm/month has been observed for cyanobacteria-dominated biocrusts common in drylands (Sorochkina et al. 2018). Field collected biocrust inoculated in a semi-cold desert had a light cyanobacteria cover of ~55% and a late successional cover (i.e., mosses and lichens) of ~14.5% in 14 months (Antoninka et al., 2018). Our study was only conducted over a single growing season with ~15% and ~25% increase in cover observed at a xeric and mesic alpine site, respectively. Interestingly, the alpine biocrusts in this study were dominated by mosses and lichens, indicating rapid propagation of these later successional communities. Increased surface moisture and higher frequency of precipitation events is likely critical to this rapid establishment in alpine environments. Our study captures only a short-term response and longer-term studies are needed to examine restoration success and community development over time.

There was no clear effect of fertilization on biocrust establishment or NA, measured via acetylene-reduction, in the experimental plots. This is contrary to Antoninka et al. 2015 who found that moss-dominated biocrusts had higher growth rates with monthly fertilization under greenhouse conditions, and (Maestre et al. 2006), who observed a positive relationship between moderate fertilizer addition and NA under laboratory conditions. Both previous studies were conducted with dryland biocrusts and nutrient limitations may not be as critical to biocrust development in alpine environments. There are contradictory results concerning nutrient addition and biocrust biomass development and nutrient status of the site may

influence the relative importance of fertilization (Maestre et al. 2006, Delgado-Baquerizo et al. 2013, Bu et al. 2015, Ayuso et al. 2017). Fertilization can result in a decrease in NA due to a negative feedback between NA and available nitrogen. The activity of symbiotic and free-living cyanobacteria are repressed at high levels of available nitrogen (Weiss et al. 2005, Gundale et al. 2013, Pérez et al. 2017). Ammonium can inhibit heterocyst differentiation and expression of *nif* genes (Liengen, 1999). Fertilization of biocrust inoculum under alpine field conditions at a rate of ~21 kg/ha N did not appear to reduce total cover in biocrust nor NA. However, we did not quantify the community composition of the inoculated biocrust and changes in species composition could occur. Not only are site differences commonly observed to interact with the biocrust fertilization treatments, but individual components of biocrust may respond differently to nutrient additions. For example, dryland biocrust moss growth was promoted with fertilization (Antoninka et al. 2015), while lichen growth in response to fertilization has been variable (Kytöviita and Crittenden, 2007) and is further complicated by varying responses of the mycobiont and photobiont within symbiosis (Johansson et al. 2011).

EPS production was not influenced by biocrust inoculation at either site. However, at Mt McIntyre, which had higher amounts of EPS, fertilization negatively influenced EPS production. EPS production is influenced by several environmental signals and regulatory pathways, including nutritional and stress conditions. Both N and Phosphate play a significant role in EPS synthesis and high levels of N can suppress the development of EPS (Mendrygal and González 2000, Janczarek 2011). It is also possible that fertilizer additions promoted decomposer populations increasing the decomposition rates of EPS in Mt McIntyre soils. Optimizing fertilization to promote EPS production under alpine field conditions requires further study.

Where the primary goal of restoration is to improve surface stability high levels of EPS production are desirable. EPS play an important role in carbon cycling in environments where soils are naturally carbon poor and EPS may provide the main input source of carbon. By stabilizing the soil, EPS reduce the carbon loss due to soil erosion (Mager and Thomas, 2011). Carbon is also the energy source cyanobacteria use to fix nitrogen. Therefore, carbon fixation is intimately linked to nitrogen fixation (Yu et al. 2014). The higher amounts of EPS present in the cold xeric alpine site may indicate that EPS plays a more important role in the carbon cycling dynamics of these soils.

Overall DOC was higher at Andesite, which is likely caused by the coastal climate's association with higher primary productivity and more carbon rich soils (Billings and Mooney 1968). The low levels of DOC at Mt McIntyre are typical of northern latitudes, where primary productivity is limited and soil development is slow (Büdel et al. 2014). At Andesite, DOC was significantly higher in inoculated plots,

suggesting that fixation and release of carbon by recovering biocrusts likely contributed to carbon stores in underlying soils. Inoculation did not appear to affect mineralizable N directly below biocrusts. A longer period of biocrust recovery may be needed to influence underlying soil N. In a degraded dryland ecosystem higher soil $\text{NH}_4\text{-N}$ was observed 18 months following inoculation with biocrust (Chiquoine et al. 2016).

Environmental conditions at the site of restoration may act as a filter altering community composition based upon the niche requirements of individual biocrust species. In a comparison between alpine and Antarctic biocrusts, Colesie et al. (2016) found that compositional difference could be correlated to climatic conditions and the adaptation necessary to thrive in extreme environments. Accordingly, in a manipulative experiment, (Li et al. 2010) found that increasing soil moisture shifted species composition to an increased proportion of mosses and lichens and a decreased proportion of cyanobacteria. In a greenhouse study, (Doherty et al. 2015) found that under higher moisture conditions liverworts and mosses may be able to rapidly colonize (Doherty et al. 2015). Although liverworts and mosses are known to host associative cyanobacteria, nitrogen fixation rates of these associations are lower than those of cyanolichens in arctic biocrusts (Hobara et al. 2006, Stewart et al. 2014). Collectively, these studies indicate that the community composition of biocrust and the ecological functioning (e.g., nitrogen fixation) of species assemblages is linked with environmental and operating conditions. Unfortunately, since we did not quantify the community composition of the recovering biocrusts this link between composition and function remains unclear. Future studies should examine how changes in the community composition of inoculated biocrusts over time are linked with recovery of biocrust function.

Conclusion

Our study provides a comprehensive examination of biocrust communities in two contrasting alpine environments. Furthermore, our work on restoration of two sites with climatic differences are a first step in understanding mechanisms driving successful restoration of biocrust in alpine environments. Although biocrust restoration was successful at both sites, Andesite showed greater recovery of nitrogen fixation, demonstrating a need for better characterization of the early successional trajectories of nitrogen fixing communities in recovering biocrust. The lower rates of nitrogen fixation following inoculation at Mt McIntyre likely indicates lower activity due to less optimal operating conditions. While restoration of key ecosystem functions is highly desirable, biocrust establishment alone could help mitigate physical impacts associated with soil surface disturbance, such as erosion. Furthermore, our study suggests that depending on restoration goals, protocols for dry and

cold climates may need to include promoting soil surface moisture retention or providing additional water inputs for successful re-establish of biocrust communities.

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